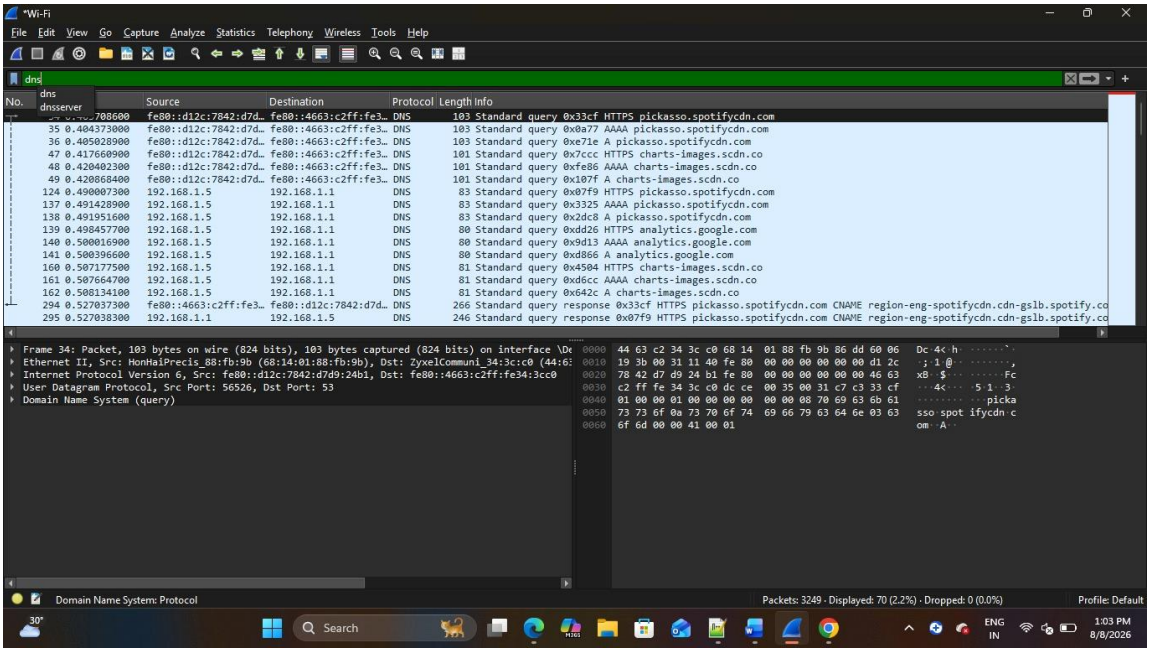
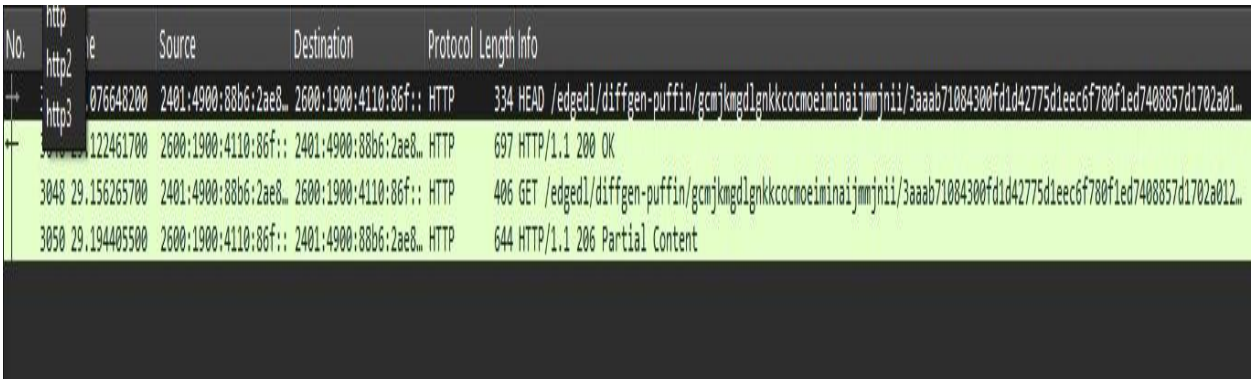
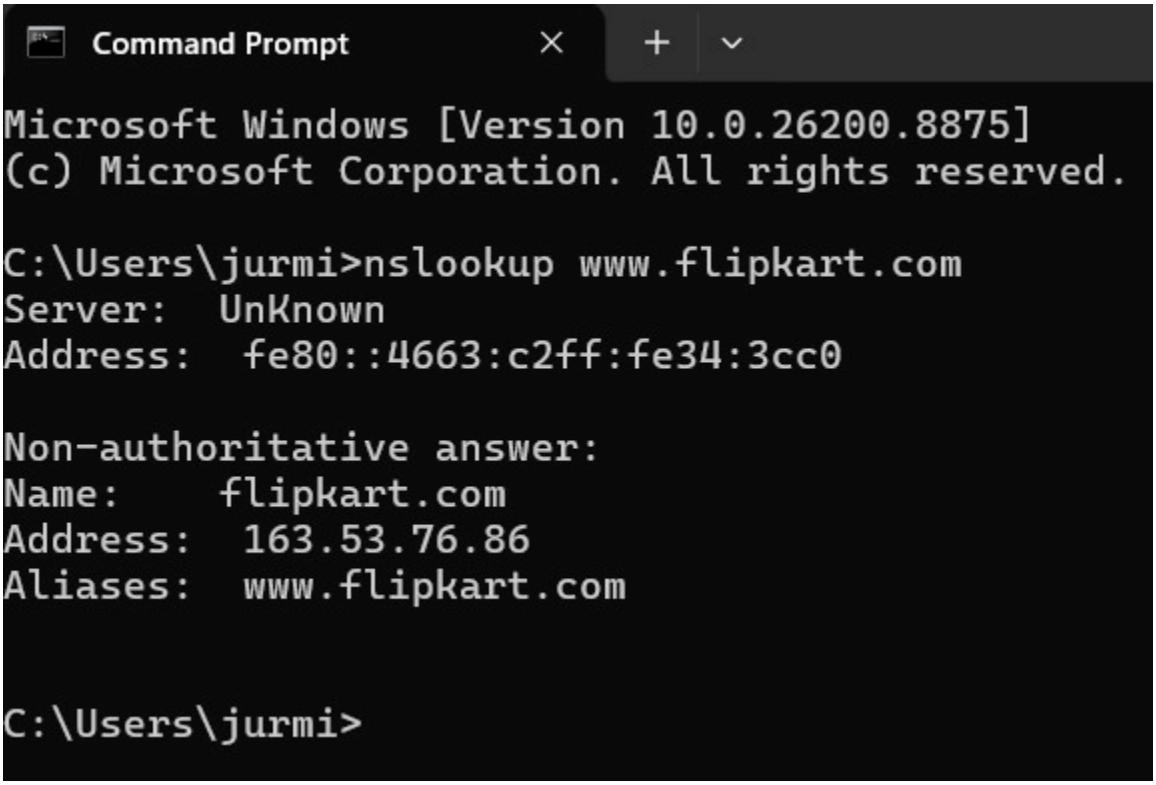
1.Use Wireshark to capture network traffic while you load www.spotify.com in your browser, then identify and note down at least two DNS packets and two HTTP or HTTPS packets in the capture.





2.Simulate a DNS lookup for www.flipkart.com using the nslookup command in your terminal and write down the IP address returned along with a brief explanation of what this result means.

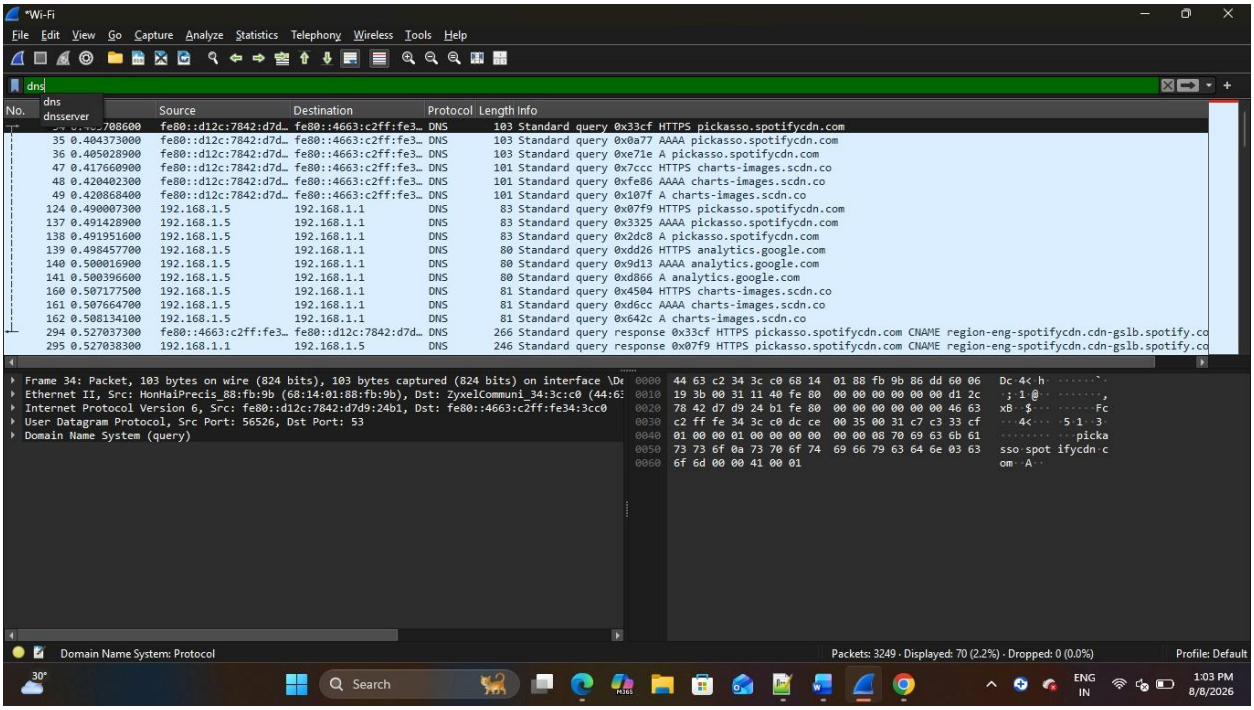


3.Explain the difference between HTTP and HTTPS by listing 3 key differences, and give a real-world example of an app or website you use that relies on HTTPS for secure communication.

Difference between HTTP and HTTPS

HTTP	HTTPS
1. Security: HTTP does not encrypt data.	HTTPS encrypts data using SSL/TLS.
2. URL: Uses http://.	Uses https://.
3. Port: Uses Port 80.	Uses Port 443.

4.Open Wireshark and filter for DNS protocol traffic. Try searching for a new website (one you haven't visited recently) in your browser and describe the sequence of DNS queries and responses you observe.Hint: Look for packets with 'Standard query' and 'Standard query response' in the Info column.



5.Pick any one protocol from DNS, DHCP, FTP, or SNMP (other than HTTP/HTTPS) and write a short scenario describing how it would be used in a real-world app or service you use (for example, how DHCP helps your phone connect to WiFi when you open Instagram at a cafe).

DHCP Protocol

When I visit a café and connect my phone to the café's WiFi, the DHCP (Dynamic Host Configuration Protocol) server in the WiFi router automatically assigns my phone an IP address, subnet mask, gateway, and DNS server. After receiving these network settings, I can open Instagram and access the internet without manually configuring any network information.